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Food mapping: a psychogeographical method for raising food consciousness

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Food mapping is a new, participatory, interdisciplinary pedagogical approach to learning about our modern food systems. This method is inspired by the Situationist International's practice of the "dérive" and draws from the discourses of critical geography, the food movement's research on food deserts, and participatory action research. Using a "critical food lens," this experiential exercise encourages participants to look beyond their plates and think about the health, economic, and ecological impacts of food. This ethnographic activity produces user-generated data and has the potential to transform participants' understanding of how agricultural practices effect other societal institutions.

Keywords: food consciousness; psychogeography; critical geography; community mapmaking; participatory action research; food movement

Introduction

Food is a simple, everyday thing, yet it is also a complex topic. Our understanding of food is influenced by historical, geographical, cultural, political, and economic factors, as well as family and personal preferences (Asp, 1999). From a holistic and ecological perspective, food is one of our species' primary energetic relationships with the planet. Many of us living in the industrialized world are removed from the sources and practice of growing food (National Institute of Food and Agriculture, 2012). While most of us are well fed and healthy, this disconnection from food production has several consequences related to our health, economic, and ecological well-being. First, most developed nations are facing a health epidemic with high incidences of obesity, heart disease, and diabetes, while consequently experiencing raising healthcare costs (Finkelstein, Trogdon, Cohen, & Dietz, 2009; Morris, 2011). Second, the globalization of our food supply is negatively affecting farmers across all social-economic, geographical, and cultural boundaries (Magdoff & Tokar, 2010; Shiva, 2005). Third, industrial agricultural methods pollute aquatic ecosystems, degrade soils, decrease biodiversity, and change our climate by generating and releasing massive amounts of greenhouse gases through deforestation, the production and use of synthetic fertilizers and pesticides, and the transportation of food over long distances (Chappell & LaValle, 2011; Dale, 2008; Forster et al., 2007; Robertson & Swinton, 2005).

Faced with these intertwined and often contested issues, policy makers, educators, and the public at large should become aware of the implications of their food choices and react accordingly. In line with *Food Web Projects* (Campaign to Protect Rural England, 1998), *Deconstructing Dinner* (Steinman, 2013), and local food indexes (Ricketts, Ilbery, &

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Kneafsey, 2006), this paper calls for increased public awareness regarding the political and economic structures dictating this unequal and ecologically exploitive agricultural system. Given the challenges of this task, we have pioneered a novel geographic, pedagogical activity called food mapping (FM) to raise citizen's food consciousness; "the degree to which an individual possesses an awareness or cognizance of decisions and actions related to food" (Giandomenico, 2000, p. 129). FM is an experiential, learner-centered exercise. Our major contributions to the literature are twofold. First, we present a synthesis of three discourses. Second, we explain the critical food lens, a master frame that encourages participants to co-produce data and conclusions about modern food systems.

To introduce our FM activity, we discuss the foundational pillars of critical geography, the food movement, and participatory action research (PAR). Then we describe the FM methodology and how to construct the critical food lens. Next, we list the necessary materials, such as data-collection sheets, informational flyers for the public, and facilitation protocols. We conclude with a discussion and evaluation regarding the benefits and challenges of this food-based critical geographic exercise.

FM foundations

This method is inspired by the Situationist International's (SI's) theory and practice of the "dérive," a psychogeographic drift through space (Debord, 1981/1958). The SI's dérive, the Flâneur, the Dadaist and Surrealist movements, the work of Walter Benjamin and, more recently, Ian Sinclair, form the theoretical background for critical geography and critical cartography when making maps (Bassett, 2004; Crampton & Krygier, 2005; Solnit, 2000). Like the dérive, FM calls upon participants to pass through or momentarily dwell in "varied ambiences" while encouraging "playful-constructive behavior," with a heightened awareness for the "psychogeographical effects" of distinctive spaces (Debord, 1981/1958, p. 50). Different food-source locations convey different feelings, information, and insights into food and our agricultural systems. Similar to the SI's vision, FM is a political activity, pushing participants to critically deconstruct their surroundings – be it fast food, a garden, or a restaurant. However, unlike the pure dérive, FM is a structured exercise. Participants question the quality, healthfulness, price, and environmental externalities associated with food production, and the spaces in which food is being sold and consumed.

The second foundation for FM comes from the international food justice movement, specifically food deserts and community mapmaking. Food deserts are geographic areas with no or distant grocery stores and limited access to affordable and nutritious food (Cummins & Macintyre, 2002; United States Department of Agriculture [USDA], 2009). Some of this research uses geographic information systems (GIS) to systematically map food sources in neighborhoods and cities (McEntee & Agyeman, 2010). Combined with census tract data, these studies have produced spatial digital pictures of the communities that do and do not have access to healthy, affordable food (Mari Gallagher Research and Consulting Group, 2007). Also drawing from critical cartography and community mapmaking, FM empowers participants to create maps that illustrate food-related issues in their communities (Crampton & Krygier, 2005; Perkins, 2007; Wood, 2010). Our FM activity offers a personalized approach to food deserts and mapmaking, inviting participants to immerse themselves in communities, restaurants, and grocery stores (or lack thereof). For example, this pedagogy prompts participants to generate maps that display community gardens, restaurants serving local organic foods, or highlight the density of fast-food outlets overlaid with health statistics and food miles associated with a purchase.

The third foundation of the FM method is PAR. PAR aims to "produce practical knowledge that is useful to people in the everyday conduct of their lives," and increase the socioeconomic, political, psychological, spiritual, and ecological well-being of people and communities (Reason & Bradbury, 2006, p. 2). PAR is also used in geographical education (Klocker, 2012; Pain, Finn, Bouveng, & Ngobe, 2013) and participatory GIS (Craig, Trevor, & Weiner, 2002). FM is a participatory, learner-centered activity in which participants generate and analyze their own data through field experiences. The data can then be represented in the form of maps, which are open to multiple interpretations, as each participant perceives food and its sources from a different point of view (Coghlan, 2005; Heron & Reason, 2006). Students and community members become co-researchers, drawing their own valid conclusions. FM meets participants where they are, creating dialogue and contributing to PAR's mission by fostering food-based "conscientization" (Freire, 1973). Finally, this pedagogy employs PAR's cyclical method of inquiry, as the activity itself is constantly evolving from participant feedback and reflection.

FM method

There is no one right way to conduct a FM exercise. This activity and these ideas are adaptable to the needs of the facilitator, participants, and context. Participants should be informed that this exercise is also designed to train facilitators for future FM sessions. Based on participant feedback, we suggest the following protocols as a guide for facilitators.

The food lens

The first and most essential part to facilitating a FM experience is developing a critical food lens (Wight, 2013), a master frame through which participants learn to analyze food sources along several different dimensions (Goffman, 1974). This conceptual tool helps participants' gain new insights into food and the space it occupies, as it asks them to consider all the processes that make our food system function. The creation of the food lens is a multi-step process and participants will reach different levels of conscientization throughout the exercise (Freire, 1973). Prior to mapping, the facilitator should pose a variety of questions about how to critically analyze food and space. These questions can be difficult to answer, and this realization in itself speaks volumes to the lack of transparency in our food systems. The prompts can be adapted to specific food products and geographical contexts depending on the goals of the facilitator and group:

(1) Where was this food grown?
(2) How far has it traveled to get to me, and which energy sources were used?
(3) Which food items are the most popular and make the most money for the producer or retailer?
(4) What are the main ingredients in most of the foods?
(5) What are the health benefits or implications associated with consumption?
(6) Does this food contain genetically modified organisms?
(7) Was this food grown using organic or conventional agricultural methods, and were chemicals, hormones, or antibiotics used to produce it?
(8) How much are the workers paid who grew, harvested, processed, and sold this food?
(9) What messages does the food space convey?

(10) What are the similarities and differences between the people who supply, prepare, and serve the food compared to those who purchase it?

Data collection and materials

Prior to mapping, the facilitator reviews and distributes the following materials: blank maps of the location, data collection sheets (Figure 1), clipboards, writing utensils, and public explanation form (Figure 2). Participants can gather data by collecting menus and food-related flyers, creating sketches of the geographic spaces, and taking pictures (Hall, 2009). The collection sheet asks the co-researchers to record the address, type of food source (restaurant, fast food, garden, gas station, grocery store, etc.), prices, nutritional, caloric, and ingredient information, hours of operation, food miles (the distance the food has traveled from farm to plate), type of production (organic or conventional), atmospherics (how the space looks, feels, and sounds), primary dishes (what is the dominate food on the menu), and brands for sale; and to write descriptive notes about their reactions to the food location and the response of the public to the activity itself. Reviewing these materials and providing examples prior to mapping can help participants construct their critical food lens and step into a psychogeographic frame.

FM protocols

We suggest a minimum of 60–90 min for the entire activity (directions, mapping, and debriefing). Working in small teams of 2–4 ensures safety, maneuverability at food locations, and provides a fun and creative atmosphere. FM can be conducted anywhere, in one food location or in several depending on the amount of time. Facilitators can predetermine food locations and ask retailers' permission beforehand, but this may detract from the participants' agency in deciding their own path (map) and the activity's spontaneity. When mapping, participants should bring smart phones, cameras, walking shoes, and appropriate clothing. They are encouraged to approach food workers and managers alike when asking questions. For beginners, FM is best conducted in urban or suburban areas, where food outlets are available. It can also be done in rural town centers or near university campuses. Next, the facilitator should review basic protocols:

Food Mapping Data Collection Sheet Name:_____Date:_____
Instructor:_____

Location check all that apply	O Grocery store O Gas station O Res hall cafeteria	O Garden O Farmers mkt O Bar	O Fast food O Street vender O Food court	O Restaurant/café O Convenience store O Other
Available options	O Dine in	O Carry out	O Drive through	O Delivery
Address Hours of Operation				
Food information for customers?	O Menu with $$	O Caloric information	O Food provider	O Other?
Notes & Prompts (be specific)	Primary dishes? Culture of food? Brands available?	Customer demographics? Worker demographics & wage? Public reactions to questions?		Atmospherics Vegan, Veggie? Organic?

Figure 1. FM data collection sheet.

Instructor Name: _____ Course Title: _____

What is going on here? Date:_____

We are mapping locations and descriptions of food. This activity is meant to create dialogue about our modern food system. We hope to raise awareness about food access, security, miles traveled, nutritional information, and the larger ecological impacts of agriculture. If you have any questions, please contact (instructor name) at (office telephone or email).

Figure 2. FM exercise public explanation.

(1) Remind participants they represent a group and should be on their best behavior.
(2) When conversing with the public, they should be open and honest about what is going on, and distribute information sheets as needed.
(3) Participants should pay attention to the passage of time and return when instructed.
(4) If they meet any resistance, simply make a note and go to the next location.
(5) When taking pictures with people, use the photo release form (Figure 3).

Follow-up discussion

After completing the FM exercise, participants meet in small groups to improve their maps, analyze pictures and data, and draw rough conclusions about their experience. First, they reflect on the activity and develop questions and impressions about the food systems. Next, the facilitator structures a larger group discussion by revisiting the critical food-lens questions, walking through the categories on the data collection sheet, and inviting feedback to see what themes emerge. It is beneficial to record all responses and hold off judgments, as each point of view is valid. We recommend creating layered, large group maps that reflect the various dimensions explored (i.e. gendered spaces, organics, healthiness scale, etc.). As participants talk about their feelings in each space, these descriptors can be written down beside the location on the class map. Finally, having been exposed to psychogeography, community mapmaking, and the critical food lens, the co-researchers are encouraged to adopt this activity for their own purposes. These future facilitators are empowered to organize future mappings and improve the data collection sheet and other materials as needed.

Course: _____ Instructor: _____

Permission to be Photographed Date: _____

I _____ (person giving permission) give _____ (person being granted permission) the right to take my picture to be used in this Food Mapping course assignment.

Figure 3. Permission to be photographed.

Evaluation and conclusion

We have conducted the FM activity six times; once at a service learning conference, another at an informal social gathering, and the rest at academic conference workshops. Participants were primarily White, college-educated individuals. A total of 44 people have participated, exploring 106 different food locations, producing 158 individual mappings (some locations were mapped by more than one person). A total of 15 rich picture maps were drawn, over 106 photographs taken, and 20 plus artifacts gathered (e.g. menus, calorie information sheets, flyers, and brochures).

Based on participant feedback, the FM exercise does indeed stimulate critical thinking about the food system, the creation of maps, and the power of geography in shaping our lives. Participants report that the critical food lens is a "strong" frame and unique approach to understanding food production and consumption. Puzzling encounters with potential food sources have left several participants wanting to know what vegetation is edible. Others spoke at length about the "dystopian" view of vending machines, fast food, and the lack of food-worker knowledge about the food being served. Many groups noted the "to-go packaging" marketed toward college students. Across groups, price was indicated as the most available piece of information. While this feedback is anecdotal and contextual, these themes suggest a heightened level of food consciousness among participants.

As a field-based, learner-centered activity, FM is a novel pedagogy for teaching about critical geography, the food justice movement, and PAR. Our major contribution to the literature is the construction and application of the critical food lens as a master frame to be used in conjunction with the dérive. This new approach gets students out of the classroom, introduces them to ethnography and qualitative data collection, and encourages them to think deeply about the global impacts of food and the representation of food spaces on maps.

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